Andres Barrios

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EDUCATION

The University of Texas at Austin May 2027
Bachelor of Science, Mechanical Engineering Honors **GPA:** 4.0

TECHNICAL SKILLS

CAD/Analysis: NX, Teamcenter, SolidWorks, Inventor, Ansys Mechanical, DFMA, Tolerance Stack-Up
Manufacturing: CNC Machining, Injection-Molding, Welding, 3D Printing, GD&T
Programming: Python, MATLAB, SQL, MongoDB, Neo4j

PROFESSIONAL EXPERIENCE

Apple – *Product Design Engineer Intern* | Cupertino, CA January 2026 - July 2026

- Led experimental research project to achieve earless eight-die cable crimping through first-principles analysis, aiming to increase crimp strength and reduce crimp envelope in future Apple cable designs
- Designed and prototyped swappable-tip crimp dies, zero-gap dies, and an adjustable cable-holding fixture in NX, using worst-case tolerance stack-ups to guarantee fit at ±0.05 mm machining tolerances
- Characterized in-house crimping machine repeatability through statistical gauge study, confirming machine error of 0.05 mm, within the 0.08 mm limit for cable jacket crimping
- Applied plastic plate buckling theory after an initial die-size sweep failed to remove ears, identifying crimp wall-thickness ratio and die gap as governing variables and redirecting testing toward thicker crimps
- Reduced ear size by 96% and increased pull force by 26% across 30 trials, synthesizing results into design rules and an axial-length tradeoff presented to guide adoption across the design team

Samsung – *Mechanical Design Engineer Intern* | Austin, TX May 2025 - August 2025

- Designed mechanical assembly in Autodesk Inventor implementing electric actuators and custom-built pipe collar to safely lift 200 lb piping within pump input subsystem, eliminating operator handling risks
- Created detailed GD&T drawings for machined pipe collar and adjustable claw clamps, using manufacturer DFM feedback to optimize parts for machining while upholding existing design constraints
- Analyzed failure modes of lifting system, confirming via hand calculations that clamping-induced radial stress remained below critical buckling stress and frictional torque resisted actuator-induced rotation

Amazon Robotics – *Hardware Development Engineer Intern* | Boston, MA January 2025 - May 2025

- Designed bolted wheel hub and cast aluminum adapter bracket in SolidWorks for new robotic drive-unit, aimed to enable wheel removal from locomotors and projected to reduce inspection downtime by over 50%
- Verified wheel assembly through tolerance stack-ups and bolted joint analysis under motor torque and lateral scrub loads, defining fastener torque spec and clearances for the prototype build
- Prototyped six configurable firmware test fixtures integrating drive unit sensors, motors, and load cells, enabling remote firmware development and exposing critical e-stop board bug during testing
- Fabricated modular test chassis to enable sensor characterization and verify field-of-vision with existing locomotors

BP – *Mechanical Reliability Engineer Intern* | Whiting, IN May 2024 - August 2024

- Developed injection quill design to minimize fluid impingement at pipe tee junction, analyzing flow conditions to optimize material selection and evaluate vibration effects on quill stability in piping system
- Designed new drainpipe for condensate removal, improving steam heat transfer efficiency to piping systems by creating new drainage line to prevent condensate accumulation in critical fluid lines

EXTRACURRICULAR EXPERIENCE

ASCEND Texas – *Mechanical Design Engineer* | Austin, TX September 2026 - Present

- Designing linear actuator mounts for parafoil control system, integrating four actuators into vehicle's 80/20 aluminum frame to actuate paraglider's B and C lines
- Measuring vehicle dimensions and working with electrical team on actuator specifications to define mount geometry, line guide placement, and strain gauge locations for critical-path subsystem